Covenant: system explanation

Covenant is a working underwriting workspace for the deal team of a fictional private credit fund, Fernhill Credit Partners. On realistic, internally consistent data it runs the full path from a new CIM to an IC-approved, locked set of terms.

1. What I built

One workspace for deals, companies and people, with each deal's work in ten tabs ordered by the deal lifecycle.

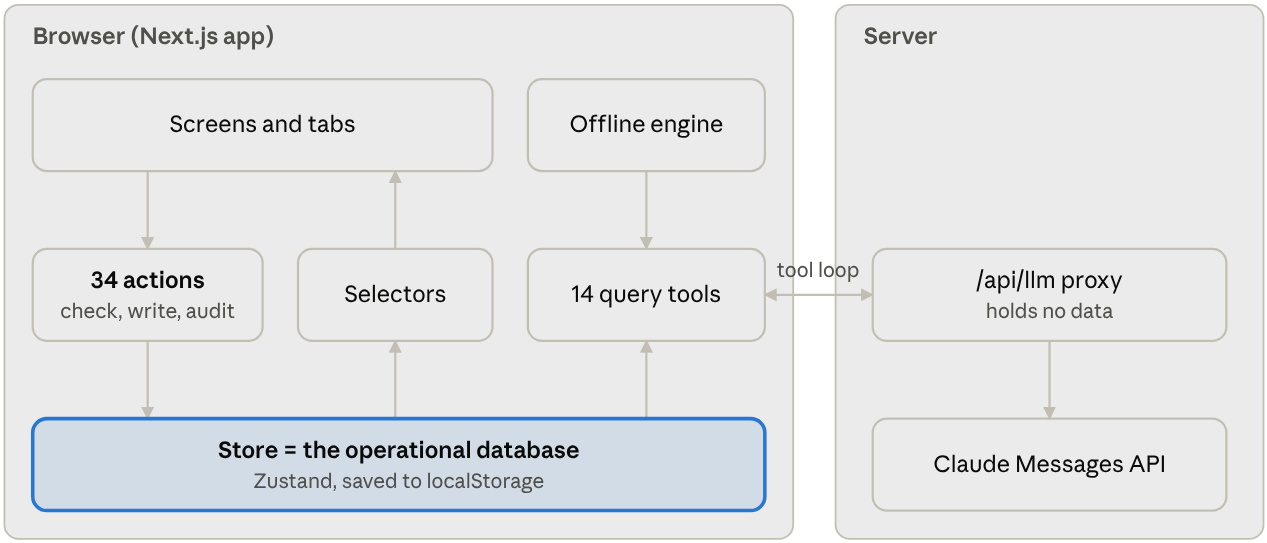
Area	What it does
Deals	List and pipeline board; dragging a deal runs the stage rules (missing items listed, overrides need a reason, Documentation needs IC approval).
Financials	AI-extracted values beside the source page, with confidence; a person confirms or corrects. Correcting LTM EBITDA from $17.5M to $16.3M moves leverage from 5.14x to 5.52x, cushion from 26.5% to 21.1%, and flags the affected memo sections.
Securities	Versioned term sets (draft, issue, compare), tranches, covenant schedule with headroom, sources and uses. IC approval locks the approved version.
Due diligence	Findings by workstream; a red flag becomes a proposed IC condition in one step, and unresolved red flags block the move to IC.
IC memo	Eight sections where every number is a live reference, never typed, so corrections can't leave stale figures. Only a VP or MD can mark it ready.
IC and approvals	Packet, votes, Q&A, conditions, decision; precedents from the fund's own past deals and devil's-advocate questions.
Chat	Questions across the whole database, answered with citations and a "Show queries" trace.
Relationships	Company and person pages with role history, sponsor track record, notes and duplicate checks.

Four personas (Analyst, VP, MD, IC member) have enforced permissions, and every change lands in an audit trail.

2. Architecture

The prototype is a Next.js app. Data lived in the browser; the server only relayed chat to Claude.

Everything ran in the browser; the server only relayed chat to Claude



Task 1 architecture

- **The data model lives in code:** schema, 34 named actions (such as `correctField`, `issueTermSet`, `decideIc`) and selectors for derived values. The UI never writes data directly.
- **AI answers by querying, not searching files:** Claude gets 14 typed query tools that return compact results with citation tokens. In production these become an MCP server over Postgres.
- **It works without AI:** with no API key, chat runs the same queries and the memo uses deterministic templates.
- **Production path:** extraction with a human review queue, Postgres, SSO and information walls, CRM and data-room integrations, and a separate analytics warehouse.

Since Task 1, the app runs on a real Postgres backend behind the same actions (see `docs/ARCHITECTURE.md`; the prototype is tagged `task1-v1`).

3. Data model

24 entities, each validated; `docs/schema.sql` is the Postgres equivalent.

- **Relationships:** Company, Person, Affiliation, Stakeholder, Note, User.
- **Underwriting:** Deal, StageEvent, Document, ExtractedField, FinancialPeriod, TermSet, Tranche, Covenant, DDItem, Memo, MemoSection, ICEvent, ICCondition, DevilsAdvocate.
- **Cross-cutting:** AuditEvent, ChatThread, ChatMessage, Settings.

Five rules hold throughout: numbers are computed from confirmed records, never typed; term sets are copy-on-write and locked at IC approval; every write produces an AuditEvent; risk ratings are stored with their scale; duplicate checks run before creating a company or person. The seed (12 deals, 25 companies, 40 people) ties out, and 69 unit tests check calculations, stage rules, permissions and seed integrity.

4. Tradeoffs

Mocked	Sign-in (persona picker); CIM extraction (staged animation with pre-written values); documents are Markdown, not PDFs.
Simplified	Browser storage as the database (one user); four roles, not deal teams; one covenant type; simple precedent similarity.
Skipped	Information walls and MNPI controls, email and calendar capture, data-room sync, Word export, closing and post-close.

5. With a couple more days

1. **Real extraction:** PDF upload, schema-driven extraction with confidence, and a firm-wide review queue.
2. **MCP server:** serve the 14 query tools over MCP, so Claude and other clients can query the fund's data directly.
3. **Financial modeling:** multi-year projections with base and downside cases, EBITDA add-backs, debt service and leverage paths over the hold, and returns (IRR, MOIC), all feeding covenant headroom; plus more covenant types.
4. **Closing:** checklist, conditions precedent, approved versus signed terms.
5. **Controls:** SSO, deal teams, information walls, Word export of the memo.